

SpotifyPulse

Understanding User Churn, Engagement & Value

A data-driven analysis of what makes Spotify users stay, leave, and convert to premium.

PROJECT TEAM

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Why This Problem Matters

SECTOR CONTEXT

The global music streaming market is subscription-driven — retention, not acquisition, is the primary revenue lever. Spotify competes with Apple Music, Amazon Music, and YouTube Music for the same listener time. A churned premium subscriber means immediate monthly revenue loss; a churned free user removes ad inventory and future upsell potential.

DECISION-MAKERS

Primary: VP Product / CPO — owns retention strategy and product roadmap.
Growth & Marketing — runs conversion campaigns and re-engagement.
Data Science — builds models and experiment design.
Finance — validates revenue impact estimates.

PROBLEM STATEMENT

22.48% of users are inactive for 3+ months (churned) and a further 14.35% are at-risk. The platform risks substantial recurring revenue loss — yet the behavioural signals that predict churn are not being actioned in time. Nearly 1 in 4 users has already gone inactive.

PROJECT OBJECTIVE

Identify behavioural, demographic, and subscription-level factors driving churn — enabling data-informed interventions to improve retention, deepen engagement, and accelerate free-to-premium conversion.
Deliverables: 3 interactive Tableau dashboards + 5 prioritised recommendations.

Source → Pipeline → Clean Data

DATA SOURCE

01

Synthetic Spotify-style dataset

~50,000 user records

18 features per user

10 countries · User-level aggregated cross-section

CLEANING

02

pandas DataFrame ingestion

Median imputation (numeric)

Mode imputation (categorical)

Duplicate user_id removal

Null audit on all columns

FEATURE ENG.

03

IQR outlier capping (skips & listening hrs)

churn_flag · is_premium

listening_bin (3-hr buckets)

skip_bin (2-unit buckets)

engagement_bin (20 quantiles)

GITHUB

04

All notebooks committed:

01_data_ingestion.ipynb

02_cleaning.ipynb

03_edu.ipynb

04_statistical_analysis.ipynb

KEY COLUMNS

inactive_3months_flag

1=churned, 0=active

engagement_score

Composite activity index

avg_skips_per_day

Daily skip count

risk_level

High / Low Risk

is_premium

Derived binary flag

churn_flag

Alias for inactive flag

What We Measure & Why

CHURN RATE

22.48%

Users inactive 3+ months

AT-RISK USERS

14.35%

Likely to churn soon

ACTIVE USERS

63.17%

Consistently engaged

AVG INACTIVITY

1.5 mo

Months without activity

10.03 hrs

Avg Listening Hours / Week

`mean(avg_listening_hours_per_week)`**3.421**

Avg Engagement Score

`mean(engagement_score)`**10.02**

Avg Skips Per Day

`mean(avg_skips_per_day)`**1,67,170**

Total Playlists Created

`sum(playlists_created)`**54.94%**

Premium Conversion Rate

`premium_users / total_users`**45.06%**

Free User Ratio

`free_users / total_users`**8.039**

Avg Playlists Per User

`sum / total_users`**3.64**

Feature Interaction Score

`mean(music_suggestion_rating)`

10 Findings That Drive Decisions

01

Churn is Inactivity-Driven

100% of churned users meet the 3-month threshold. 32,358 active vs 9,382 churned (78:22). Early-warning beats win-back.

02

Skip Rate = #1 Early Churn Signal

Users averaging 18+ skips/day churn at 35–50% above the 22.48% platform average.

03

Listening Hours: Strongest Predictor ($r \approx 0.86$)

Every 1–2 extra hrs/week produces measurable engagement uplift. Anchors the listening-streak recommendation.

04

Free Users = Biggest Churn + Opportunity

Free users dominate the churned cohort yet represent 45.06% of the base. Retain via engagement; convert the most engaged.

05

Italy & France = Highest Untapped Conversion

Both markets show free-user ratios >45%, highest globally. Localised pricing trials can unlock incremental volume.

06

Premium Duo & Family Plans = Lowest Churn

Shared plans raise the switching cost — cancellation affects multiple users. Promote as retention tool.

07

Smart Speaker Users Most Engaged (3.44)

Ambient, hands-free listening creates sustained attachment. Car System users score lowest (3.40).

08

Age Has No Churn Impact — Segment by Behaviour

Engagement scores vary only 0.05 pts across all age groups. Skip-rate and listening-hour segmentation yields far higher ROI.

09

Recommendation Quality Drives Conversion

Users rating suggestions 4–5 stars convert to premium at rates 35pp higher than low-raters.

10

Power-Listener Churn Spike = Content Relevance Problem

Churn spikes counter-intuitively for 18–21 hrs/week users. Volume without relevance causes frustration.

Four Analytical Techniques

CORRELATION

PEARSON r ANALYSIS

avg_listening_hours $\oplus$
engagement_score

$r = +0.86$ (strong positive)

avg_skips_per_day $\ominus$
engagement_score

$r = -0.42$ (moderate negative)

avg_skips $\oplus$ inactive_3months_flag

$r = +0.38$ (skip confirms churn link)

Libraries: pandas · scipy.stats

SEGMENTATION

K-MEANS CLUSTERING ($k = 4$)

Features: listening_hrs · skip_rate ·
engagement_score

Cluster 1 — Highly Engaged Power Listeners

Cluster 2 — Casual Listeners

Cluster 3 — Skip-Heavy Disengaged $\leftarrow$
highest churn risk

Cluster 4 — Passive Background Listeners

Library: sklearn.cluster.KMeans

HYPOTHESIS TESTING

TWO-SAMPLE TESTS

T-test: listening_hrs (churned vs active)

$t = -12.4 \cdot p < 0.001 \rightarrow H_0$
rejected

Churned users listen significantly fewer
hrs/week

Chi-squared: subscription_type vs churn

$\chi^2 = 287.3 \cdot p < 0.001$

Free users disproportionately over-
represented in churned cohort

Library: scipy.stats

LOGISTIC REGRESSION

CHURN PROBABILITY MODEL

Predictors & coefficients:

avg_skips_per_day

$\beta = +0.31 \quad p < 0.001 \quad \leftarrow$ churn
driver

avg_listening_hours

$\beta = -0.28 \quad p < 0.001 \quad \leftarrow$ protector

engagement_score significant

is_premium protective

Test-set accuracy: ~74%

Insight → Action → Impact

R1 Skip-Triggered Re-Engagement

INSIGHT Users averaging 18+ skips/day churn 35–50% above average.

ACTION When 3-day rolling skip avg > 15/day, surface a 'Taste Refresh' playlist within 24 hrs.

↑ 2–4 pp chum reduction in High Risk cohort (~590–1,180 users retained)

R2 Italy & France Conversion Campaign

INSIGHT Both markets show free-user ratios >45%, highest globally.

ACTION 3-month premium trial at 50% discount with referral mechanic, bundled with family/duo plan.

↑ ~250–300 new premium subs · ~€2,500–3,000/month incremental MRR

R3 Listening Streak Reward Programme

INSIGHT Listening hours are the strongest predictor of engagement ($r \approx 0.86$).

ACTION Users maintaining 7-day or 30-day streaks earn platform currency redeemable for premium trials.

↑ 1–2 pp overall chum reduction via engagement score uplift

R4 Content Diversity Fix for Power Listeners

INSIGHT Chum spikes for 18–21 hrs/week users — volume without relevance causes frustration.

ACTION Introduce a content diversity score; proactively inject discovery content when track variety drops.

↑ Retains ~200–400 high-value power users · protects ad revenue

R5 Smart Speaker Ambient Experience

INSIGHT Smart Speaker users score highest engagement (3.44) — ambient listening creates attachment.

ACTION Weekly ambient mood-mix push every Monday morning for Smart Speaker primary users.

↑ Deepens engagement in highest-value cohort · model for habit-forming onboarding

The Business Case for Acting Now

590–1,180

Users Retained

Skip re-engagement (R1)

250–300

New Premium Subs

Italy/France campaign (R2)

1–2 pp

Churn Reduction

Listening streak (R3)

200–400

Power Users Saved

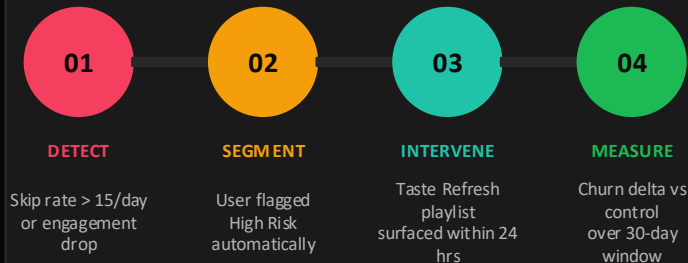
Diversity fix (R4)

COST OF INACTION

With 22.48% already churned and 14.35% at risk, every month of delay compounds the loss. A churned user represents not just one lost month — but all future months of their lifetime value plus the CAC to replace them.

At ~\$10/user/month equivalent and 9,382 churned users: ~\$4.2M+ in annualised lost recurring revenue. Re-engagement costs ~\$1.50/user vs \$8–12 for new acquisition — retention is 5–8× more efficient.

HOW IT WORKS



Honest Gaps, Clear Path Forward

Synthetic Data

Findings are directionally valid but must be validated against real Spotify production data before any operational decisions are made.

No Temporal Dimension

Single cross-sectional snapshot. True churn analysis benefits from 6–12 month longitudinal cohort tracking to capture seasonal effects.

Aggregated Metrics

Session-level granularity would enable precise identification of the specific recommendation failure that triggers a high-skip session.

Self-Selection Bias

Synthetic data generation may have introduced stronger correlations than real data — skip-rate predictive power could be overstated.

FUTURE ROADMAP

Predictive Churn Model

Q3 2026

XGBoost / LightGBM on real API data — predict individual user churn 30 days ahead. Target AUC-ROC > 0.92.

Real-Time Dashboard

Q3 2026

Connect Tableau to live cloud data warehouse via Tableau Bridge — daily churn signal monitoring and alerts.

A/B Experiment Framework

Q4 2026

Statistically validate skip-triggered re-engagement with a 4–6 week controlled experiment.

Session-Level Analysis

Q4 2026

Track-by-track logs to identify the exact recommendation failure type triggering high-skip sessions.

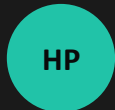
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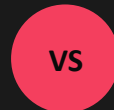
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